

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester I)

Scala For DS-9

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Class :FYIT	Batch : 2
Date of Assignment :21-07-2026	Date/Time of Submission :21/07/26

Aim:Find the correlation between two lists of numbers. Implement the formula for person correlation coefficient

Code:

```
import breeze.linalg._
object Prac9 extends App {
    val x = List(1.0,2.0,3.0,4.0,5.0)
    val y = List(2.0,4.0,6.0,8.0,10.0)

    val meanX = x.sum / x.length
    val meanY = y.sum / y.length

    var num = 0.0

    for (i <- 0 until x.length) {
        num = num + (x(i) - meanX) * (y(i) - meanY)
    }

    var denX= 0.0
    var denY= 0.0

    for(i<- 0 until x.length){
        denX = denX + (x(i) - meanX) * (x(i) - meanX)
    }
    for(i<- 0 until x.length){
        denY = denY + (y(i) - meanY) * (y(i) - meanY)
    }

    val den = math.sqrt(denX * denY)
    val cor = num / den
    println("Person correlation coefficient : " + cor )

    println("Himanshu S037")
}
```

Output:

```
"C:\Program Files\Java\jdk1.8.0_341\bin\java.exe" ...
```

```
Person correlation coefficient : 1.0
```

```
Himanshu S037
```

```
Process finished with exit code 0
```